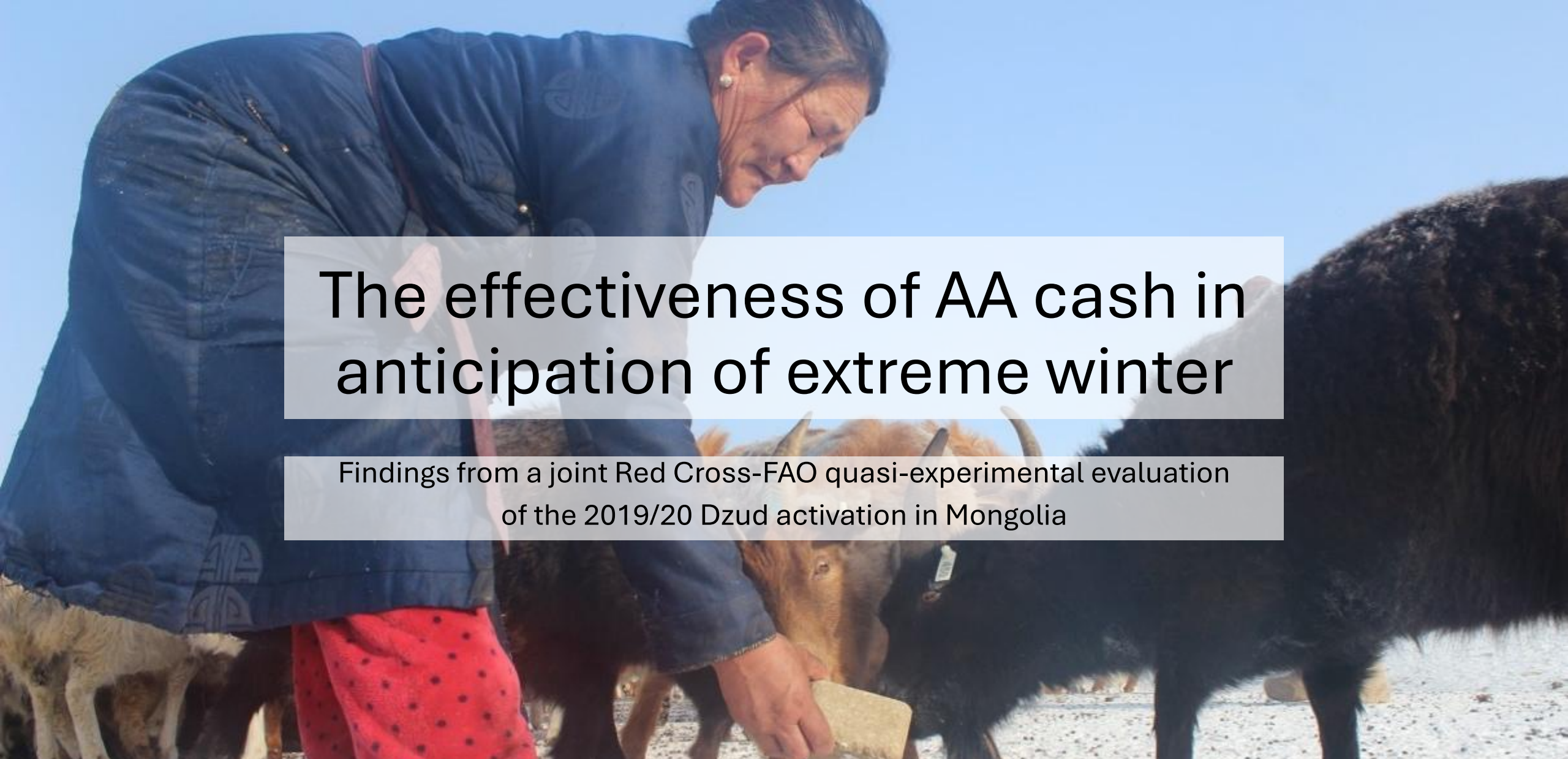


A woman in a blue jacket and red skirt is feeding a goat in Mongolia. The background is a clear blue sky.

The effectiveness of AA cash in anticipation of extreme winter

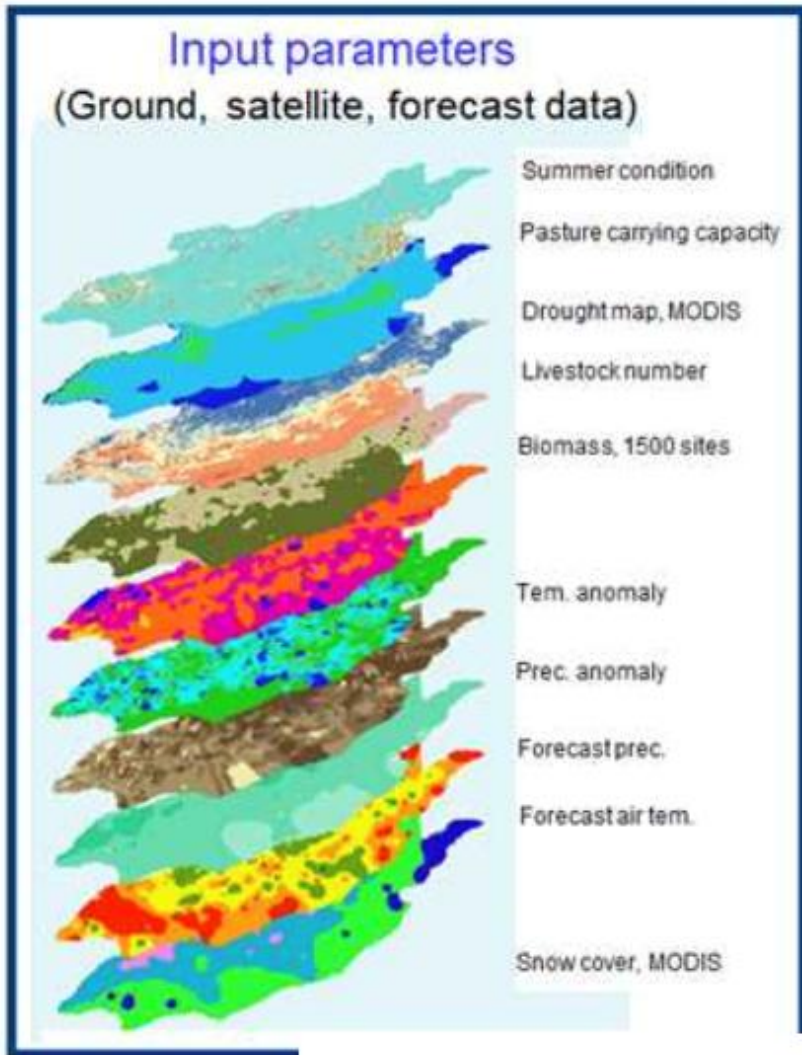
Findings from a joint Red Cross-FAO quasi-experimental evaluation of the 2019/20 Dzud activation in Mongolia



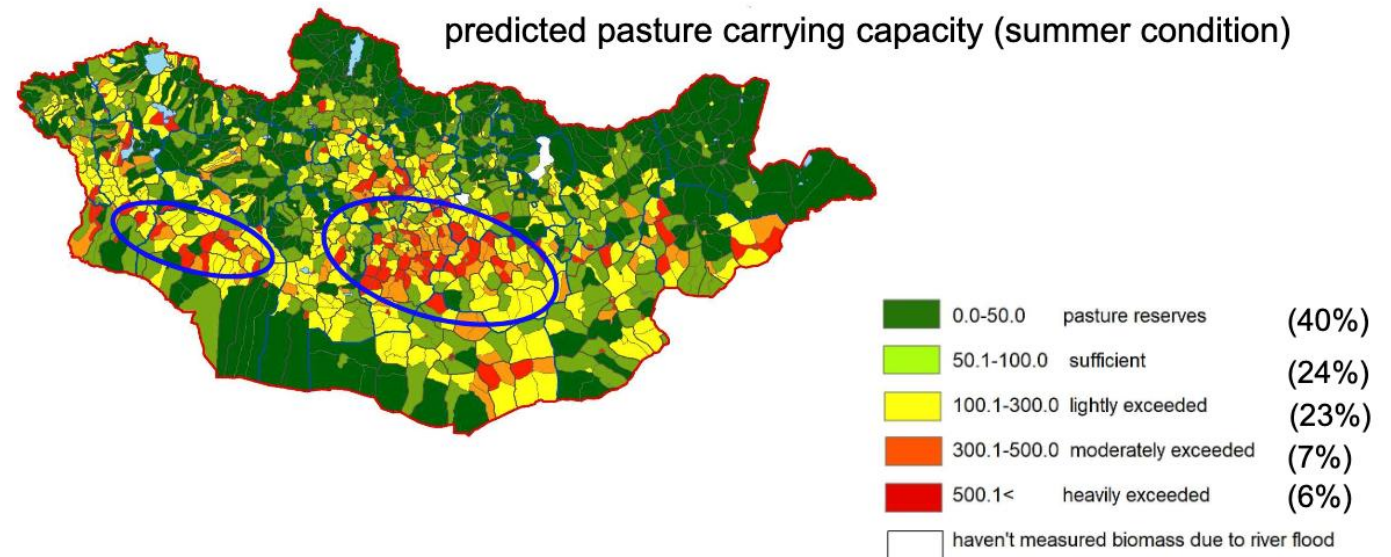
Unusual weather (extreme **cold**) and/or land surface conditions (lack of pasture and deep **snow/ice cover**; limited/no access to dried grasses under deep snow) that prevent livestock pasture accessibility & availability, resulting **massive livestock loss** during winter-spring.

Pastoralist livelihoods **at risk from Dzud**

Every fall, a Dzud risk map is issued > **Nov. 2019:**



Winter Outlook for 2019/2020 based on carrying capacity and current condition (snow depth)



Low Dzud risk in most of Mongolia.
BUT: **Black Dzud** could occur in some areas.

2nd Jan 2020: Severe Dzud risk warning issued

National Agency for Meteorology and Environmental Monitoring (NAMEM): 50% of the country at risk of an extreme winter.

- By 12th Jan, 70% of the country was covered by up to **30cm of snow**.
- In some regions **local hay and fodder reserves** had been **completely depleted** at soum level.



Cash transfers & in-kind assistance: From **anticipation** to **response**



Unconditional cash grant
(US\$ 84-96 equiv.)



Animal care kit

Three interventions

	Red Cross Anticipatory Action	FAO Anticipatory Action	Red Cross RESPONSE
First actions reaching beneficiaries	27 January 2020	3 March 2020	18 March 2020
Actions completed	5 February 2020	14 March 2020	17 April 2020
# of beneficiaries	1,000 households	450 households	1,750 households
Geographic coverage	8 provinces, 61 soums	5 provinces, 22 soums	15 provinces, 82 soums
Cash transfer per beneficiary household	240,000 MNT (~ 84 US\$)	275,000 MNT (~ 96 US\$)	240,000 MNT (~ 84 US\$)
Other actions:	Livestock care kit: •30 kg of mineral block •2 litres of fish oil •1 kg of livestock vitamin •1 piece of hoof oil •1 piece of eye ointment	Livestock care kit: •2 litres of fish oil •2 kg of livestock vitamin •1 piece of hoof oil •1 piece of eye ointment Bag for new-born kids and lambs (booklet + feed additive)	Livestock care kit: •30 kg of mineral block •2 litres of fish oil •1 kg of livestock vitamin •1 piece of hoof oil •1 piece of eye ointment

Evaluation objective & methods

- **Objective:** Assess the effects of providing anticipatory cash and livestock care kits at different times on the livelihoods and well-being of herder households experiencing Dzud conditions
- **Quasi-experimental design**, balancing the trade-off between rigour (sample size) and operational/financial feasibility
- **Estimation approach:** To avoid attrition in our already small sample, we use inverse probability weighting (IPW) regressions after propensity score matching (PSM) to control for the differences between the groups when assessing the effects of the anticipatory and response interventions on the ability of Mongolian herder households to cope with the impacts of dzud

Sample composition and background characteristics

Socio-demographic characteristics	RC AA	FAO AA	RC Resp.	Control
Education of the respondent: completed secondary school (% n)	55.94 (113)	64.00* (96)	58.71 (118)	55.22 (111)
Number of elderly HH members >= 60 (mean, se)	0.27** (0.04)	0.09 (0.03)	0.22 (0.04)	0.15 (0.03)
Number of disabled household members (mean, se)	0.36*** (0.04)	0.15 (0.03)	0.28** (0.03)	0.18 (0.03)
Households with any additional source of income (% n)	95.54 (193)	86.00*** (129)	94.53 (190)	96.52 (194)
Distance of winter shelter from soum centre in km (mean, se)	29.30*** (1.86)	31.75*** (1.80)	39.50*** (1.79)	16.85 (1.00)
Households received an early warning on dzud (% n)	44.06*** (89)	44.00*** (66)	49.25*** (99)	21.39 (43)
Sample size:	202	150	201	201

Note: Comparisons are shown between intervention (MRCS, FAO, DREF) and control households using chi-square tests (categorical variables) and independent sample t-tests (continuous variables). Means and standard errors in parenthesis are reported for continuous variables; percentage (%) and number of observations (n) in parenthesis for categorical variables. Statistically significant are shown at the level of 1%***, 5%** and 10%*.

Summary of evaluation results (1): Destitution sales & borrowing

AA/response effect & size against the comparison group (no assistance)

Outcome variables	MRCS AA	FAO AA	MRCS Resp.
Amount of debt per capita taken (mean, MNT)	 -377,147 MNT (132 US\$)	•	•
Whether sold any assets to cope with the impacts of dzud (%)	 -17%	 -43%	 -20%
Share of livestock in sheep head unit (SHU) sold to cope with the impact of dzud (%)	 +8%	•	•

No effect on beneficiaries' need to borrow money to cope with Dzud impacts.
(21-26% of beneficiaries took out new loans; average loan size 600,000-750,000 MNT).

Visual representation of regression coefficients showing the average treatment effect applying inverse probability weighting (IPW) regression following propensity score matching to control for differences in socio-demographic background characteristics. Robust standard errors are reported in parentheses below coefficient estimates. Statistical significance for each coefficient is indicated at the level of 1%***, 5%** and 10%*.

Summary of evaluation results (2): Livestock health

AA/response effect direction & size against the comparison group (no assistance)

Outcome variables	MRCS AA	FAO AA	MRCS Resp.
Share of animals in SHU with deteriorated body condition (proxy of morbidity %)	 -7%	 -13%	
Offspring survival rate during Dzud (%)	 5.3%		

No effect on animal mortality in Sheep Head Units (SHU).

Comparison by individual species – without controlling for differences between groups – shows the following significant differences:

- **MRCS** AA: horse mortality (-3.3%)
- **FAO** AA: cattle (-5.1%) and sheep (-3.7%) mortality

Summary of evaluation results (3): Consumption, expenditure & income

AA/response effect & size against the comparison group (no assistance)

Outcome variables	MRCS AA	FAO AA	MRCS Resp.
Monthly per capita household income from all sources (mean, MNT)	 31,190 MNT (11 US\$)		 19,522 MNT (7 US\$)
Whether COVID-19 affected household's ability to cope with the impacts of dzud (%)			 -13%
Food consumption score (FCS) (mean)		 -3.16	 3.48
Whether reduced household consumption expenditure due to the impacts of dzud on income (%)	 18%		

Conclusion

- All three interventions – Red Cross (MRCS) and FAO anticipatory actions, MRCS response – were **highly effective** in helping vulnerable herder households **avoid destitution sales** of valuable assets to cope with the impacts of Dzud, reducing this coping behaviour by 17% (MRCS) to as much as 43% (FAO).
- Both **AA** interventions were effective in helping herders reduce the share of sheep with **weakened body conditions** by 13% (FAO) and 7% (MRCS).
- Mongolian context **makes robust research design** relatively **costly** (large intervention area; low population density; high mobility)

